So far for the good news: you can also see that there are a few points that are not close to any of the spherical clusters: we added some noise to the dataset as well, to show one of k-means critical issue: it can't detect outliers, and in fact, as shown in figure 12.5, outlier points are added to the closest clusters. Since centroids are computed as centers of mass of the clusters, and the mean function is sensitive to outliers, if we don't filter out outliers before running k-means, the undesirable consequence is that the centroids of the clusters will be "attracted" by outliers away from the best position they could hold – you can see this phenomenon in several clusters in figure 12.5.

The issue with outliers, however, is not the worst problem with k-means. As mentioned, this algorithm can only produce spherical clusters, but unfortunately in real datasets not all clusters are spherical! Figure 12.6 shows three examples where k-means clustering fails to recognize the best clustering.

Not-linearly separable clusters can't be approximated with spherical clusters, and as such k-means can't separate non-convex clusters like (clusters shaped as) two concentric rings; moreover, in all those situations where the clusters' shape is not spherical and the points can't be separated correctly using minimum distance from a centroid, k-means will struggle to find good solutions.

Another issue with k-means is that the number of clusters is a hyper-parameter, meaning that the algorithm is not able to determine the right number of clusters automatically[7], and instead it takes the number of centroids as an argument: this means that, unless we have some insight deriving from